SET: 1

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AGRA REGION
FIRST PRE-BOARD EXAM 2024-25
CLASS: XII
SUBJECT- PHYSICS(THEORY)

Max. Marks : 70 **Time: 3 hours**

General Instructions:

(1) There are **33 questions** in all. All questions are compulsory.

(2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.

(3) All the sections are compulsory.

(4) **Section A** contains **sixteen** questions, **twelve MCQ** and **four Assertion Reasoning** based of **1 mark each**, **Section B** contains **five** questions of **two marks** each, **Section C** contains **seven questions** of **three marks** each, **Section D** contains two case study-based questions of **four marks** each and **Section E** contains **three long answer** questions of **five marks each.**

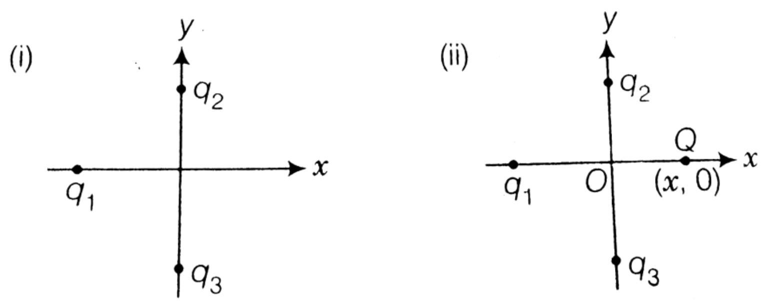
(5) There is no overall choice. However, an internal choice has been provided in **one question** in

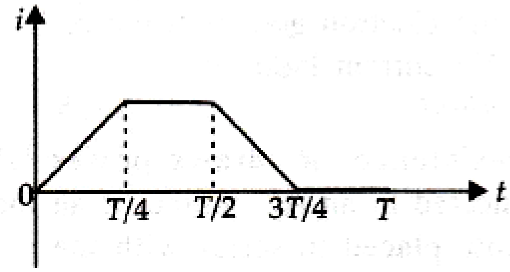
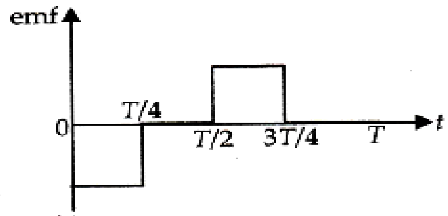
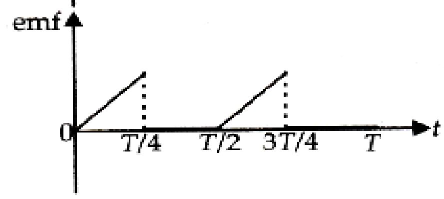
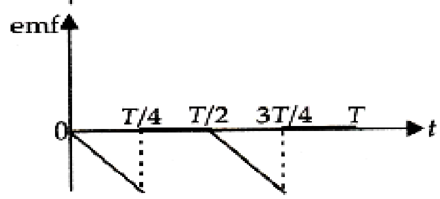
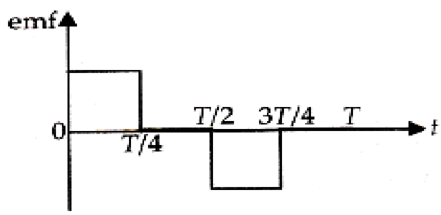
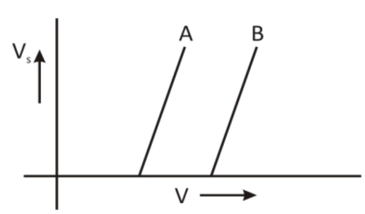
Section B, **one question** in Section C, one question in each CBQ in Section D and all three

questions in Section E. You have to attempt only one of the choices in such questions.

(6) Use of calculators is not allowed.
(7) **You may use the following values of physical constants where ever necessary**
i. $c = 3 \times 10^{8}$ m/s
ii. $m_e = 9.1 \times 10^{-31}$ kg
iii. $e = 1.6 \times 10^{-19}$ C
iv. $\mu_0 = 4\pi \times 10^{-7}$ TmA^{-1}
v. $h = 6.63 \times 10^{-34}$ Js
i. $\varepsilon_0 = 8.854 \times 10^{-12}$ $C^2N^{-1}m^{-2}$
vii. Avogadro's number = 6.023×10^{23} per gram mole

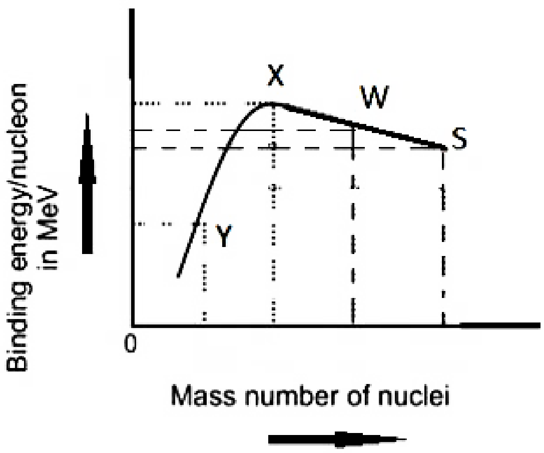
	(SECTION-A) (16x1=16)	
Q.NO.	QUESTIONS	Marks
01	In a Wheatstone bridge, all the four arms have equal resistance R. if resistance of the galvanometer arm is also R, then equivalent resistance of the combination is (a) R (b) 2R (c) R/2 (d) R/4	1
02	When a ferromagnetic material is heated above the curie temperature it becomes: (a) Diamagnetic (b) Paramagnetic (c) Strongly charged (d) Non-Magnetic	1
03	One requires 11 eV of energy to dissociate a carbon monoxide molecule into carbon and oxygen atoms. The minimum frequency of the appropriate electromagnetic radiation to achieve the dissociation lies in (a) visible region (b) infrared region (c) ultraviolet region (d) microwave region	1
04	In an experiment of single slit diffraction, the width of a slit 1.2µm and the angular width of centre maximum is observed to be equal to $\boldsymbol{\pi}$ **/3** find the wavelength of light (a) 6 A^0 (b) 60 A^0 (c) 600A^0 (d) 6000A^0	1
05	A nucleus of mass number 189 splits into two nuclei having mass no125 and 64.the ratio of radius of daughter nuclei respective is (a) 25:16 (b) 1:1 (c) 4:5 (d) 5:4	1
06	The electrostatic potential on the surface of a charged conducting sphere is 100V. Two statements are made in this regard S_1 at any point inside the sphere, electric intensity is zero. S_2 at any point inside the sphere, the electrostatic potential is 100V. Which of the following is a correct statement?	1

	(a) S_1 is true but S_2 is false (b) Both S_1 and S_2 are false (c) S_1 is true, S_2 is also true and S_1 is the cause of S_2 (d) S_1 is true, S_2 is also true but the statements are independent	
07	In figure two positive charges q_2 and q_3 fixed along the y-axis, exert a net electric force in the +x-direction on a charge q_1, fixed along the x-axis. If a positive charge Q is added at (x, 0), the force on q_1  (a) shall increase along the positive x-axis (b) shall decrease along the positive x-axis (c) shall point along the negative x-axis (d) shall increase but the direction changes because of the intersection of Q with q_2 and q_3	1
08	A galvanometer coil has a resistance of 12 **Ω** and the metre shows full scale deflection for a current of 3 mA. How will you convert the metre into a voltmeter of range 0 to 18 V? (a) 5888 ohm in series (b) 5888 ohm in parallel c) 5988 ohm in series (d) 5988 ohms in parallel	**1**
09	The current I in a coil varies with time as shown in figure below.	1

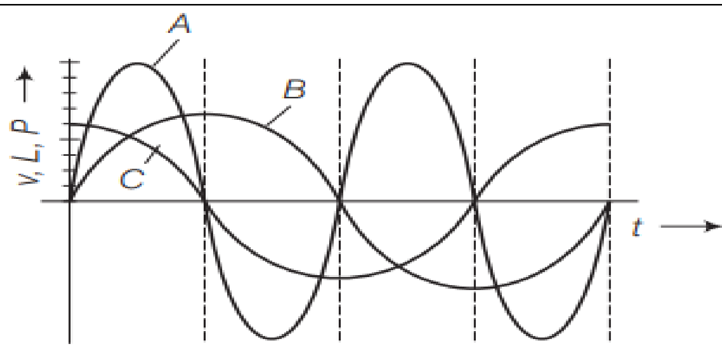
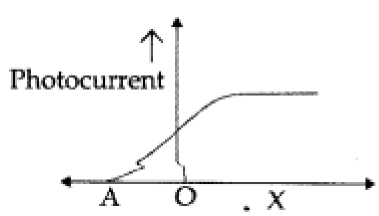
	 The variation of induced e.m.f. with time would be: (a)  (b)  (c)  (d)  	
10	The stopping potential as a work function of frequency of incident radiation is plotted for two different photo electric surfaces A and B. The graphs show the work function of A is  (a) greater than that of B	1

	(b) smaller than that of B (c) same as that of B (d) no comparison can be done from given graph	
11	In an interference pattern produced by two coherent sources of light, the position of the dark fringe corresponds to: (a) Zero path difference (b) Quarter wavelength path difference (c) Wavelength path difference (d) Odd multiple of half-wavelength path difference	1
12	If refractive index of water is 5/3. A light source is placed in water at a depth of 4 m. Then what must be the minimum radius of disc placed on water surface so that the light of source can be stopped? (a). 5m (b). 4m (c). 3m (d). Infinity	1
	For Questions 13 to 16, two statements are given –one labeled **Assertion (A)** and other labeled **Reason (R).** Select the correct answer to these questions from the options as given below. A. If both Assertion and Reason are true and Reason is the correct explanation of Assertion. B. If both Assertion and Reason are true but Reason is not the correct explanation of Assertion. C. If Assertion is true but Reason is false. D. If both Assertion and Reason are false.	
13	**Assertion (A):** A charged capacitor is disconnected from a battery. Now, if its plates are separated further, the potential energy will fall. **Reason (R):** Energy stored in a capacitor is not equal to the work done in charging it	1

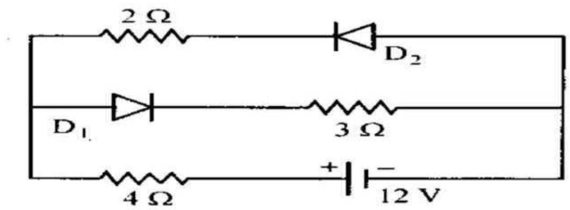
14	**Assertion (A):** A bulb connected in series with a solenoid is connected to ac source. If a soft iron core is introduced in the solenoid, the bulb will glow brighter. R**eason((R)** : On introducing soft iron core in the solenoid, the inductance decreases.	1
15	**Assertion (A):** In Young's double-slit experiment, if one of the slits is covered, the interference pattern disappears. **Reason (R):** Interference occurs due to the superposition of waves from two coherent sources	1
16	**Assertion(A):** The depletion layer in p-n junction under forward bias decreases. **Reason(R):** The electric field due to external voltage supports the electric field due to potential barrier.	1
	[SECTION – B] (05x2=10 marks)	
17	A cylindrical conductor of length l and cross-section area A is connected to a DC source. Under the influence of electric field set up due to source, the free electrons begin to drift in the opposite direction of the electric field. (I)Draw the curve showing the dependency of drift velocity on relaxation time. (II) If the DC source is replaced by a source whose current changes its magnitude with time such that $I = Io \sin 2\pi\nu t$, where ν is the frequency of variation of current, then determine the average drift velocity of the free electrons over one complete cycle.	1+1
18	Two long parallel straight wires X and Y separated by a distance of 5 cm in air carry current of 10 A and 5 A respectively in opposite directions.	1.5+0.5

	Calculate the magnitude and direction of the force on a 20 cm length of the wire Y.	
19	A single slit of width 0.1 mm is illuminated by a parallel beam of light of wavelength 6000 A and diffraction bands are observed on a screen 0.5 m from the slit. Find the distance of the third dark band from the central bright band .	2
20	Binding energy per nucleon versus mass number curve is as shown. X,Y,W and S are four nuclei indicated on the curve.  Based on the graph (a) Arrange X, W and S in the increasing order of stability. (b) Explain why binding energy for heavy nuclei is low.	1+1
21	Draw the circuit diagram of full wave rectifier and also draw its input and output wave forms. **OR** How is forward biasing different from reverse biasing in a p-n junction diode?	1+1/2 +1/2 **OR** 2
	[SECTION – C] **(07x3=21 marks)**	

22	a) Use Gauss's theorem to find the electric field due to a uniformly charged infinitely large plane thin sheet with surface charge density (b) An infinitely large plane thin sheet has a uniform surface charge density. Obtain the expression for the amount of work done in bringing a point charge q from infinite to a point, distance d , in front of the charged plane sheet. OR State Gauss's theorem in electrostatics. Using this theorem, derive an expression for the electric field due to an infinitely long straight wire of linear charge density λ.	1.5+1.5 Or 1+2
23	A cell of emf 'E' and internal resistance 'r' is connected across a variable load resistor R. Draw the plots of the terminal voltage V versus (i) R (ii) the current I. (iii) It is found that when R = 4 Ω, the current is 1 A and when R is increased to 9 Ω, the current reduces to 0.5 A. Find the values of the emf E and internal resistance r.	1+1+1
24	A bar magnet of magnetic moment 1.5 J T^{-1} lies aligned with the direction of a uniform magnetic field of 0.22 T (a) What is the torque on the magnet when magnetic moment aligned (i) Normal to the field direction (ii) Opposite to the field direction (b) Write the conditions for (i) stable equilibrium (ii) unstable equilibrium of magnet.	1+1+ 1/2+ 1/2
25	A device 'X' is connected to an AC source. The variation of voltage, current and power in one complete cycle is shown in figure.	

	 (a) Which curve shows power consumption over a full cycle? (b) What is the average power consumption over a cycle? (c) Identify the device X.	1 1 1
26	Name the e.m. waves in the wavelength range 10 nm to 10^{-3} nm. How are these waves generated? Write their two uses.	1+1+1
27	Draw a ray diagram to show the formation of the image of an object placed on the axis of a convex refracting surface, of radius of curvature 'R', separating the two media of refractive indices 'n_1' and 'n_2' ($n_2 > n_1$). Use this diagram to deduce the relation $\frac{n2}{v} - \frac{n1}{u} = \frac{n2-n1}{R}$, where u and v represent respectively the distance of the object and the image formed.	3
28	The given graph shows the variation of photocurrent for a photosensitive metal:  (a) Identify the variable X on the horizontal axis. (b) What does the point A on the horizontal axis represent? (c) Draw this graph for three different values of frequencies of incident radiation v_1 v_2 and v_3 ($v_1 > v_2 > v_3$) for same intensity.	1/2+1/2+1+1

	(d) Draw this graph for three different values of intensities of incident radiation I_1 I_2 and I_3 ($I_1 > I_2 > I_3$) having same frequency.	
	SECTION D (02x4=08 marks)	
29	**Case Study Based Question: Semiconductor** There are different techniques of fabrication of p-n junction. In one such technique, called fused junction techniques, an aluminum film is kept on the wafer of n-type semiconductor and the combination is then heated to a high temperature (about 600 °C). As a result, aluminum fused into silicon and produces p-type semiconductor and in this way p-n junction is formed. **(i) when a PN junction is reversed then how does the height of potential barrier change?** (a) no current flows (b) the depletion region is reduced (c) height of potential barrier is decreased (d) height of potential barrier is increased **(ii) the cause of potential barrier in PN junction is:** (a) depletion of positive charges near the junction (b) concentration of negative charges near the junction (c) concentration of positive and negative charges near the junction (d) depletion of negative charges near the junction **(iii) The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit ?**	1x4



(a) 1.17 A (b) 2.0 A (c) 1.71 A (d) 1.

(iv) Carbon, Germanium and silicon are 14th group elements:

(a) C and Ge are semiconductors (b) C and Si are semiconductor

c) all C, Ge and Si are semiconductors (d) Si and Ge are semiconductors

Or

When a PN junction is forward biased then:

(a) only diffusion current flows

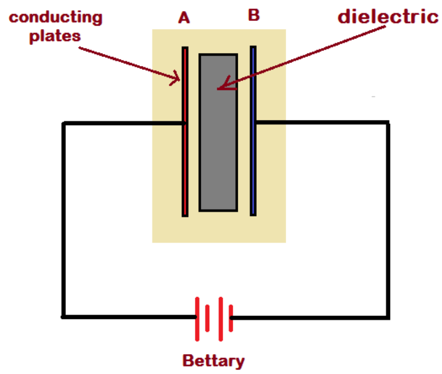
(b) both diffusion current and drift current flow but diffusion current is more than drift current

(c) only drift current flows

(d) both diffusion and drift current flow but drift current exceeds the diffusion current.

30 Case Study Based Question: Electrostatics 1x4

An arrangement of two conductors separated by an insulating medium can be used to store electric charge and electric energy. Such a system is called a capacitor.



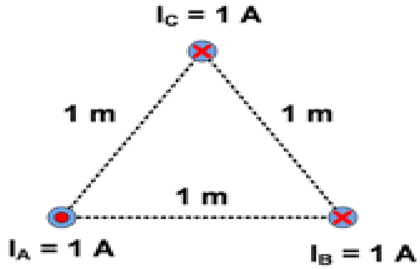
The more charge a capacitor can store, the greater is its capacitance. Usually, a capacitor consists of two conductors having equal and opposite charge +Q and -Q. Hence, there is a potential difference V between them. By the capacitance of a capacitor, we mean the ratio of the charge Q to the potential difference V. By the charge on a capacitor we mean only the charge Q on the positive plate. Total charge of the capacitor is zero. The capacitance of a capacitor is a constant and depends on geometric factors, such as the shapes, sizes and relative positions of the two conductors, and the nature of the medium between them. The unit of capacitance is farad (F), but the more convenient units are μF and pF. A commonly used capacitor consists of two long strips or metal foils, separated by two long strips of dielectrics, rolled up into a small cylinder. Common dielectric materials are plastics (such as polyesters and polycarbonates) and aluminium oxide. Capacitors are widely used in radio, television, computer, and other electric circuits.

1. A parallel plate capacitor C has a charge Q . The actual charge on its plates are

(a) Q,Q (b) Q/2, Q/2 (c) Q ,-Q (d) Q/2, -Q/2

2. A parallel plate capacitor is charged. If the plates are pulled apart .

	(a) the capacitance increases (b) the potential difference increases (c) the total charge increases (d) the charge & potential difference remains the same 3. Three capacitors of 2 , 3 & 6 μF are connected in series to a 10 V source. The charge on the 3 μF capacitor is (a) 5μC (b) 10μC (c) 12μC (d) 15μC 4. If n capacitors each of capacitance C are connected in series and then in parallel, then the ratio of equivalent capacitance of the series combination to parallel combination is (a) nC b) n^2C (c) C/n (d) $1/n^2$ **OR** A parallel plate capacitor has two square plates with equal and opposite charges. The surface charge densities on the plates are + σ and - σ respectively. In the region between the plates the magnitude of the electric field is (a) σ / ε_0 (b) $\sigma / 2 \varepsilon_0$ (c) 0 (d) none of these	
	SECTION E **03x5= (15 marks)**	
31	(i) Three conductors form an equilateral triangle shown the figure below. The current direction of wire A is out of the plane of the figure, in B and C are	3+ 2

	 into the plane of the figure. Find: a) the magnetic force acts on conductor B in a length of 1 m b) the direction of the force (ii) Write the mathematical expression for the force acting on a current carrying straight conductor kept in a magnetic field. Under what conditions is this force (i) zero and (ii) maximum? **OR** With the help of a neat and labelled diagram, explain the underlying principle and working of a moving coil galvanometer. What is the function of (i) uniform radial field (ii) soft iron core in such a device?	Or 1+ 1+ 1+ 1+ 1
32	(a)Draw a ray diagram to show refraction of a ray of monochromatic light passing through a glass prism. Deduce the expression for the refractive index of glass in terms of angle of prism and angle of minimum deviation. (b) A convex lens has 20 cm focal length in air. What is focal length in water? (Refractive index of air-water = 1.33, refractive index for air-glass = 1.5.)	1+2+2

	OR (a) Draw a labelled ray diagram to obtain the real image formed by an astronomical telescope in normal adjustment position. Define its magnifying power. (b) You are given three lenses of power 0.5 D, 4 D and 10 D to design a telescope. (i) Which lenses should be used as objective and eyepiece? Justify your answer. (ii) Why is the aperture of the objective preferred to be large?	Or 1+1+2 +1
33	(i)Using Bohr's postulates of the atomic model, derive the expression for radius of nth electron orbit. Hence obtain the expression for Bohr's radius. (ii) According to the classical electromagnetic theory, calculate the initial frequency of the light emitted by the electron revolving around a proton in hydrogen atom. OR (i) In Rutherford scattering experiment, draw the trajectory traced by a-particles in the Coulomb field of target nucleus and explain how this led to estimate the size of the nucleus. (ii) In a Geiger-Marsden experiment, what is the distance of closest approach to the nucleus of a 7.7 MeV α-particle before it comes momentarily to rest and reverses its direction?	3+2 Or 3+2

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